From Time Crystals to Thermal Phases beyond Symmetry Breaking

Equilibrium order with and without symmetry breaking

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This talk is based on

I. H. Watanabe, M. Oshikawa, PRL 114, 251603 (2015); H. Watanabe, M. Oshikawa, T. Koma, J. Stat. Phys. 178, 926 (2020)

II. S. Watanabe, Y. Motome, H. Watanabe, PRB 114, 094108 (2026); arXiv:2603.03852; PRX 16, 041016 (2026); JPSJ 95, 085002 (2026)

III. H. Watanabe, arXiv:2607.00134; S. Sengoku, W. Cao, H. Watanabe, arXiv:2610.27938; G. Xun, W. Cao, H. Watanabe, arXiv:2610.27939

Order in thermal equilibrium: Landau's paradigm and two questions at its edges

Setting: a short-range $\hat{H}$ on a lattice; equilibrium = its ground state or its Gibbs state $\rho_\beta = e^{-\beta\hat{H}}/Z$.
Landau: phases $\leftrightarrow$ patterns of spontaneous symmetry breaking, diagnosed with no symmetry-breaking field by *long-range order* of a local $\hat{\phi}$: $\langle\hat{\phi}(\boldsymbol{x})\hat{\phi}(\boldsymbol{x}')\rangle \to \sigma^2 \neq 0$ as $|\boldsymbol{x}-\boldsymbol{x}'| \to \infty$ (after $V \to \infty$).

Question I: can *every* symmetry break?

Wilczek (2012): a **time crystal** would break *time* translation as a crystal breaks space translation: $\langle\hat{\phi}(\boldsymbol{x},t)\hat{\phi}(0,0)\rangle \to f(t)$ oscillating in t.

Part I: no — for short-range interactions

Question II: is symmetry *all* there is?

Can two Gibbs states at $T > 0$ be **different phases** with no local order parameter telling them apart, as topological order does at $T = 0$?

Parts II–III: water: yes, symmetry suffices; 3D toric code in fields: no

Two questions, one object: which long-range correlations can a short-range $\hat{H}$ sustain in equilibrium? Locality answers both.

Landau, Zh. Eksp. Teor. Fiz. 7, 19 (1937); Wilczek, PRL 109, 160401 (2012); H. Watanabe & M. Oshikawa, PRL 114, 251603 (2015); H. Watanabe, arXiv:2607.00134

The common thread: which correlations survive the thermodynamic limit?

Every verdict in this talk is about a correlation function of a short-range $\hat{H}$ in equilibrium as $V \to \infty$: no symmetry-breaking field, no hand-picked order parameter.

	correlation	$V \to \infty$	verdict
symmetry breaking (Landau)	local $\hat{\phi}$ at **equal times**: $\langle \hat{\phi}(\boldsymbol{x}) \hat{\phi}(\boldsymbol{x}') \rangle$	$\to \sigma^2 \neq 0$	yes: crystals, magnets, ice VIII
time crystal (Part I)	local $\hat{\phi}$ at **different times**: $\langle \hat{\phi}(\boldsymbol{x}, t) \hat{\phi}(0, 0) \rangle$	$\to f(t)$ non-constant	**no**: no-go theorem at $T = 0$ and $T > 0$
loop order in ice (Part II)	global flux P of closed spin loops (dual XY/Ising order)	quantized only at $v = 0$	**no at** $T > 0$: monopoles open the loops $\Rightarrow$ crossover
beyond symmetry (Part III)	**nonlocal**: two winding Wilson loops after local decoding, $\langle W_1 W_2 \rangle_c$	$\to f_W \neq 0$	**yes**: 3D toric code in fields

Order is a correlation that survives the thermodynamic limit. Part I asks whether it can live in *time*; Parts II–III ask whether it can live in *nonlocal* observables at $T > 0$.

Plan of the talk

I. **Time crystals.** Can continuous time-translation symmetry break in equilibrium?
a definition by long-range order in time, and a no-go theorem for short-range interactions

II. **Beyond symmetry at $T > 0$?** What it would mean; then ice VII, VIII, X and spin ice, where symmetry turns out to be enough
Chern insulator at $T > 0$, two-way local channels; monopoles open the loops $\Rightarrow$ crossover

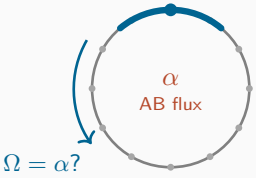
III. **3D toric code in magnetic fields.** A thermal phase beyond symmetry
closed flux loops $\Rightarrow$ an invariant $\mathcal{F}$ of two-way local-channel phases;
theorems for classical $\mathbb{Z}_2$ cell models

Part I: Time crystals — can time translation break in equilibrium?

Wilczek's proposal: can time-translation symmetry break spontaneously?

A **time crystal** breaks continuous *time* translation as a crystal breaks space translation: $\langle \hat{O}(t) \rangle$ oscillates in the ground state.

- **Wilczek (2012)**: N bosons on a ring with an Aharonov–Bohm flux α, mean field $i\partial_t \psi = \left[\frac{1}{2}(-i\partial_x + \alpha)^2 - \lambda|\psi|^2\right]\psi$
- attraction $\lambda > \lambda_c = \frac{\pi}{2}(1 - 4\alpha^2)$: the ground state is a soliton; with $\alpha \neq 0$ the solution $\psi \propto \psi_0(x - \alpha t)$ rotates, so the current is periodic in t
- **Bruno (2013)**: it is not the lowest-energy state; the true one, $\psi = e^{-i\mu t}\phi(x)$, is stationary, and in the rotating frame $E_0^{(\Omega)} - E_0^{(0)} > 0$ for $\Omega \neq 0$
- a general verdict needs a general definition: eigenstates trivially give time-independent $\langle n|\hat{O}(t)|n\rangle$, yet ordinary crystals do exist



bosons on a ring with flux α:
does the soliton rotate forever?

Can time translation break spontaneously in equilibrium? First, a definition.

Wilczek, PRL 109, 160401 (2012); Shapere & Wilczek, PRL 109, 160402 (2012); Li *et al.*, PRL 109, 163001 (2012); Bruno, PRL 111, 070402 (2013); H. Watanabe, Süri Kagaku 62, 45 (2024)

Defining time crystals: long-range order in time

Ordinary long-range order is defined by correlations alone, in the *symmetric* ground state or Gibbs ensemble with no symmetry-breaking field: $\langle\hat{\phi}(\boldsymbol{x})\hat{\phi}(\boldsymbol{x}')\rangle \to \sigma^2 \neq 0$ at large distance, i.e. $\lim_{V\to\infty}\langle\hat{\Phi}^2\rangle/V^2 = \sigma^2$ for $\hat{\Phi} := \int_V d^dx\,\hat{\phi}(\boldsymbol{x})$.

Time crystal (Watanabe–Oshikawa 2015): long-range order *in time*, i.e. correlation between distant points at *different* times,

$$\lim_{V\to\infty}\langle\hat{\phi}(\boldsymbol{x},t)\hat{\phi}(0,0)\rangle \to f(t) \quad \text{for large } |\boldsymbol{x}|, \qquad f(t) \text{ non-constant (periodic) in } t$$

equivalently, for the integrated $\hat{\Phi}$: $\lim_{V\to\infty}\langle e^{i\hat{H}t}\hat{\Phi}e^{-i\hat{H}t}\,\hat{\Phi}\rangle/V^2 = f(t)$

- a time-dependent symmetry-breaking field would leave the energy defined only modulo the drive frequency
- independent precessing spins oscillate on one site but have no long-range correlation: not a crystal
- space-time crystal: $f_{\boldsymbol{G}}(t)$ for $\hat{\Phi}_{\boldsymbol{G}} = \int d^dx\,\hat{\phi}(\boldsymbol{x})e^{-i\boldsymbol{G}\cdot\boldsymbol{x}}$; $\hat{\phi} = \hat{\rho}$ generalizes crystalline order

A time crystal is long-range order in time: $f(t)$ must stay nontrivial as $V \to \infty$.

No time crystals at $T = 0$

Theorem (ground state)

For short-range interactions and extensive $\hat{A} = \int_V d^d x\, \hat{a}(\boldsymbol{x})$, $\hat{B} = \int_V d^d x\, \hat{b}(\boldsymbol{x})$ with local $\hat{a}, \hat{b}$,

$$\frac{1}{V^2}\left|\langle 0|\hat{A}\, e^{-i(\hat{H}-E_0)t}\,\hat{B}|0\rangle - \langle 0|\hat{A}\hat{B}|0\rangle\right| \le C\,\frac{t}{V} \qquad (C \text{ independent of } t, V)$$

Proof. For $\Delta(t) := \langle 0|\hat{A}e^{-i(\hat{H}-E_0)t}\hat{B}|0\rangle - \langle 0|\hat{A}\hat{B}|0\rangle$, integrate in time and use Schwarz:

$$\begin{aligned}|\Delta(t)| &\le \int_0^t ds\, \left|\langle 0|\hat{A}(\hat{H}-E_0)e^{-i(\hat{H}-E_0)s}\hat{B}|0\rangle\right| \\ &\le \tfrac{t}{2}\sqrt{\langle 0|[\hat{A},[\hat{H},\hat{A}]]|0\rangle\,\langle 0|[\hat{B},[\hat{H},\hat{B}]]|0\rangle} = O(t\,V) \qquad \square\end{aligned}$$

- each commutator removes a factor of V: $\|[\hat{A},[\hat{H},\hat{A}]]\| = O(V)$, so $f(t) = f(0)$ at every fixed t
- power law $r^{-\alpha}$, $0 < \alpha < d$: bound $C\,t\,V^{-\alpha/d} \to 0$; space-time crystals: Hermitian parts of $\hat{\Phi}_G$

$\langle[\hat{\Phi},[\hat{H},\hat{\Phi}]]\rangle = O(V) \ll V^2$: locality alone forbids long-range order in time at $T = 0$.

No time crystals at $T > 0$: Gibbs states

For excited states $\sqrt{\hat{H} - E_n}$ is undefined, so use locality as the **Lieb–Robinson bound**:

$$\left\| [e^{i\hat{H}t}\hat{a}(\boldsymbol{x})e^{-i\hat{H}t}, \hat{b}(\boldsymbol{y})] \right\| \leq \min\{C_1,\, C_2 e^{-\mu(|\boldsymbol{x}-\boldsymbol{y}|-vt)}\}$$

- $\Rightarrow g_{AB}(t) \coloneqq \langle [\hat{A}(t), \hat{B}] \rangle_\beta / V^2 = O\big((1 + (vt)^d)/V\big) \to 0$ for $t = o(V^{1/d})$
- spectral representation $f_{AB}(t) = \int d\omega\, \rho_{AB}(\omega) e^{-i\omega t}$, $g_{AB}(t) = \int d\omega\, (1 - e^{-\beta\omega}) \rho_{AB}(\omega) e^{-i\omega t}$: a vanishing commutator pushes the spectral weight to $\omega = 0$, so $f_{AB}(t)$ is constant (PRL 2015, sketch)
- complete proof (Watanabe, Oshikawa, Koma 2020): $\hat{H} = \sum_x \hat{h}_x$ bounded, finite-range, $\hat{A} = |\Lambda|^{-1} \sum_x \hat{a}_x$; smooth frequency cutoffs and four energy-difference windows give $|\langle \hat{A}(t)\hat{A}^\dagger \rangle_\beta - \langle \hat{A}\hat{A}^\dagger \rangle_\beta| \leq 4\epsilon + 2C/K^2 + 2N_a^2 \epsilon |t|$, hence $\lim_{|\Lambda|\to\infty} \big(\langle \hat{A}(t)\hat{B} \rangle_\beta - \langle \hat{A}\hat{B} \rangle_\beta \big) = 0$ at every $\beta > 0$; extends to a class of stationary states

No-go theorem: with short-range interactions, continuous time translation never breaks spontaneously in equilibrium, at $T = 0$ or $T > 0$. Time is not another direction of space: the spectrum of $\hat{H}$ is bounded below.

H. Watanabe & M. Oshikawa, PRL 114, 251603 (2015); H. Watanabe, M. Oshikawa, T. Koma, J. Stat. Phys. 178, 926 (2020); Lieb & Robinson, CMP 28, 251 (1972)

Part I: summary

- Time crystals, defined by long-range order in time ($\langle\hat{\phi}(\boldsymbol{x},t)\hat{\phi}(0,0)\rangle \to f(t)$ non-constant as $V \to \infty$), do not exist in ground states or Gibbs states of systems with short-range interactions
- What the theorem leaves open is nonequilibrium: Floquet (discrete) time crystals stabilized by many-body localization or prethermalization, and dissipative open systems — none of them thermal equilibrium
- The lesson for equilibrium: the Gibbs state is built from $\hat{H}$ itself, whose spectrum is bounded below; time is not just another direction of space

Equilibrium cannot break time translation. Which distinctions between thermal states **can** exist beyond symmetry breaking? $\Rightarrow$ Parts II and III

reviews: Else, Monroe, Nayak, Yao, Annu. Rev. Condens. Matter Phys. 11, 467 (2020); Khemani, Moessner, Sondhi, arXiv:1910.10745; Sacha & Zakrzewski, Rep. Prog. Phys. 81, 016401 (2018)

Part II: Beyond symmetry at $T > 0$? — and water, where symmetry is enough

What distinguishes two phases of matter?

At $T = 0$: classification goes beyond symmetry

- intrinsic topological order
- symmetry-protected topological phases
- quantized responses

At $T > 0$: the Landau paradigm

- phases $\leftrightarrow$ patterns of spontaneous symmetry breaking
- local order parameters
- same symmetry $\Rightarrow$ connectable?

Do Gibbs states $\rho = e^{-\beta H}/Z$ at $T > 0$ admit a classification **beyond symmetry breaking**?

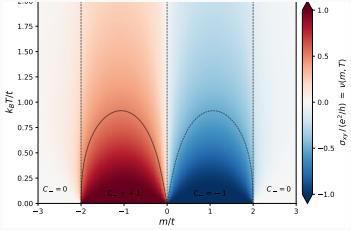
Landau, Zh. Eksp. Teor. Fiz. 7, 19 (1937); Wen, Int. J. Mod. Phys. B 4, 239 (1990); Kitaev, Ann. Phys. 303, 2 (2003); Chen, Gu, Liu, Wen, PRB 87, 155114 (2013)

Example: a Chern insulator at $T > 0$ loses its topological distinction

Qi–Wu–Zhang model (two-band Chern insulator): $h_{\boldsymbol{k}} = \boldsymbol{g}_{\boldsymbol{k}} \cdot \boldsymbol{\sigma}$, $\boldsymbol{g}_{\boldsymbol{k}} = (\sin k_x,\ \sin k_y,\ m - \cos k_x - \cos k_y)$

$\sigma_{xy}(T) = \frac{e^2}{\hbar A} \sum_{\boldsymbol{k}} \sum_n f(\epsilon_{n\boldsymbol{k}})\, \Omega_n(\boldsymbol{k})$, $\quad \Omega_n(\boldsymbol{k}) = -2\,\mathrm{Im}\langle \partial_{k_x} u_{n\boldsymbol{k}} | \partial_{k_y} u_{n\boldsymbol{k}} \rangle$ (Kubo; f: Fermi function)

- $T \to 0$: $\sigma_{xy} = \frac{e^2}{h} C_-$ with $C_- = -\mathrm{sgn}(m)$ for $0 < |m| < 2$ and $C_- = 0$ for $|m| > 2$: distinct phases separated by gap closings
- $T > 0$: σ_{xy} is no longer quantized and the free energy is analytic in m: Chern and trivial insulators are **connected**
- SPT order protected by an on-site symmetry also cannot persist at $T > 0$; with a 1-form symmetry it can

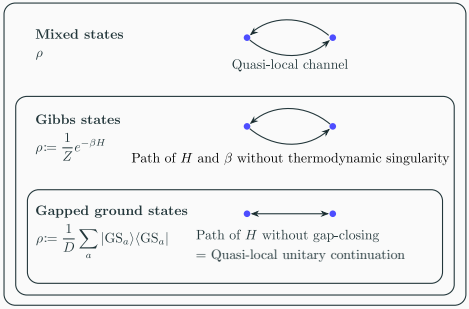


$\nu(m, T) = \sigma_{xy}(T)/(e^2/h)$ at $\mu = 0$; dashed: $T = 0$ gap closings

Ground-state topological invariants do not label thermal states.
Goal: a topological invariant that distinguishes phases of Gibbs states at $T > 0$.

What does “the same phase” mean for a mixed state?

- gapped ground states: local unitary / quasi-adiabatic continuation
- general mixed states: **two-way local quantum channels** — $\rho \sim \sigma$ iff local channels with product ancillas and causal radius $o(L)$ convert $\rho \to \sigma$ **and** $\sigma \to \rho$ with trace-norm error $\to 0$
- Gibbs states: paths without thermodynamic singularity



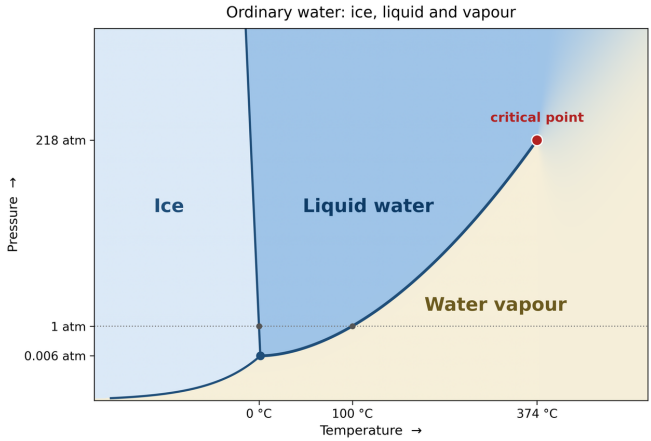
The classification of mixed-state phases is still **developing**, with several competing definitions. Whether the two-way local-channel definition and free-energy analyticity single out the same phases is **not known**.

Hastings & Wen, PRB 72, 045141 (2005); Chen, Gu, Wen, PRB 82, 155138 (2010); Coser & Pérez-García, Quantum 3, 174 (2019); Sang, Zou, Hsieh, PRX 14, 031044 (2024)

The first phase diagram we ever learn

- Ice, liquid water, vapour
- The liquid–vapour line **ends** at a critical point ($374\,^{\circ}$C, 218 atm)
- Beyond it, liquid and vapour transform into each other **continuously**

Liquid water and steam are **one** thermodynamic phase.



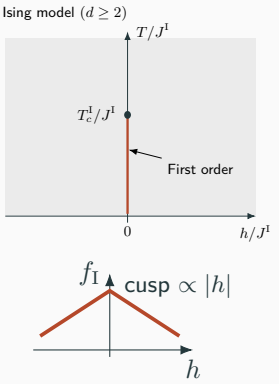
Stanley, *Introduction to Phase Transitions and Critical Phenomena* (1971); Chaikin & Lubensky, *Principles of Condensed Matter Physics* (1995)

The Ising lesson: the liquid–vapour line of a magnet

Ferromagnetic Ising model on a $d \geq 2$ lattice, $\sigma_i = \pm 1$:

$$H_{\mathrm{I}} = -J^{\mathrm{I}} \sum_{\langle ij \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i, \qquad J^{\mathrm{I}} > 0$$

- cusp at $h = 0$: $f_{\mathrm{I}}(T, h) = f_{\mathrm{I}}(T, 0) - m_{\mathrm{sp}}(T)|h| + o(|h|)$ for $T < T_c^{\mathrm{I}}$ — a first-order line ending at a critical point
- lattice gas ($\sigma_i = +1$: occupied, $h \leftrightarrow$ chemical potential): this *is* the liquid–vapour line; water lacks the $\mathbb{Z}_2$ symmetry, so its line bends, but it also ends
- the ordered phase is $\mathbb{Z}_2$-protected and lives only on the symmetric line: any $h \neq 0$ connects the two sides smoothly



Symmetry pins the boundary: the ordered phase lives on the symmetric line $h = 0$ and nowhere else; off it, nothing distinguishes the two sides.

Why can the liquid–vapour line end, but not the melting line?

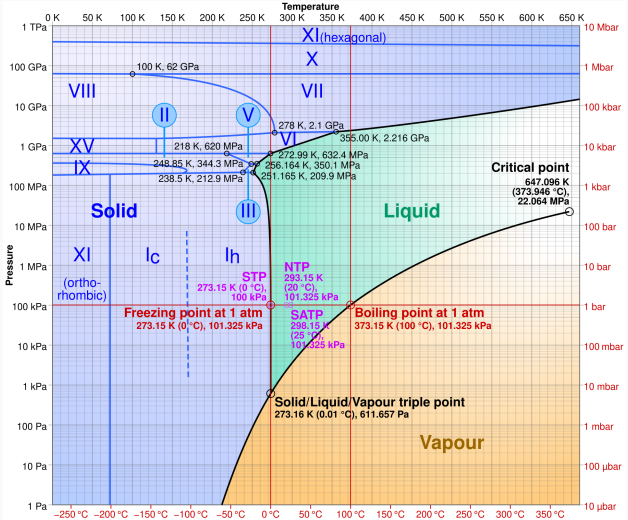
	broken symmetry	local order parameter
vapour	none	—
liquid	none	—
ice	translations, rotations	crystalline density wave

- Two states with the **same** symmetry may be connected without a singularity
- Two states with **different** symmetry are always separated by a phase boundary

Landau–Ginzburg: at $T > 0$, phases are classified by **spontaneous symmetry breaking**. Is this complete?

Landau (1937); Anderson, *Basic Notions of Condensed Matter Physics* (1984)

The phase diagram of water



Cmglee, "Phase diagram of water," Wikimedia Commons, CC BY-SA 3.0 (log–lin chart, 1 Pa–1 TPa, 0–660 K; data compiled from Chaplin, *Water Structure and Science*, lsbu.ac.uk/water)

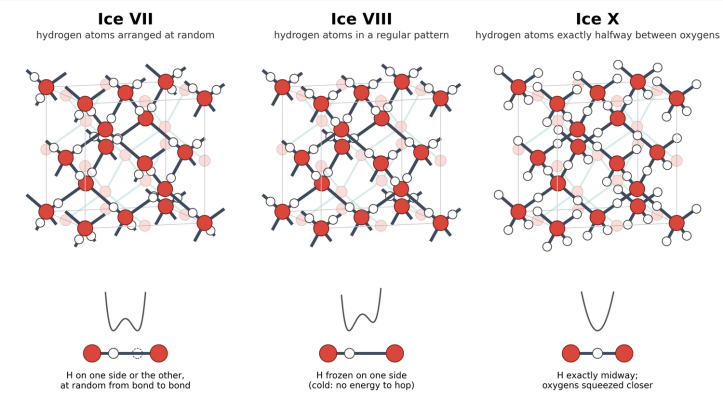
The icefields: more than twenty crystalline phases

- Ice I_h, I_c, II, III, ..., XX: from negative pressure to > 100 GPa
- Almost all are distinguished by their **space group** $\Rightarrow$ the Landau classification applies
- Exceptions that share one space group: **ice VII and ice X** ($Pn\bar{3}m$), ice III and IX ($P4_12_12$)

Are ice VII and ice X different phases, or the same phase in different regimes?

Bartels-Rausch *et al.*, Rev. Mod. Phys. 84, 885 (2012); Salzmann *et al.*, Phys. Chem. Chem. Phys. 13, 18468 (2011); Hansen, Nat. Commun. 12, 3161 (2021)

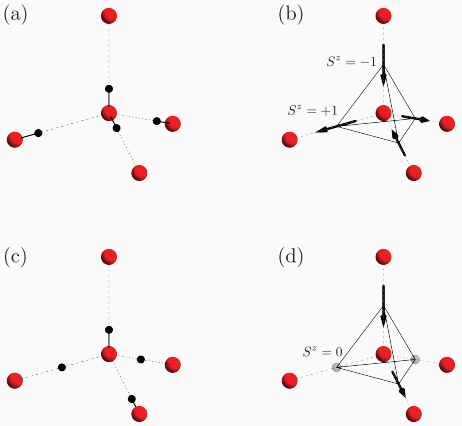
Dense ice: one oxygen framework, three hydrogen arrangements



- **VII**: protons disordered on two off-center sites per bond (ice rules) — $Pn\bar{3}m$
- **VIII**: protons ordered, antiferroelectric — $I4_1/amd$, cubic symmetry broken
- **X**: protons at the bond midpoints — $Pn\bar{3}m$, **same space group as VII**

Kuhs *et al.*, J. Chem. Phys. 81, 3612 (1984); Hemley *et al.*, Nature 330, 737 (1987); Holzapfel, J. Chem. Phys. 56, 712 (1972)

From protons to spins: a spin-1 pyrochlore magnet



- O–O bond midpoints form a **pyrochlore lattice**
- Proton near A / at midpoint / near B $\iff S^z_\ell = -1, 0, +1$
- Ice rules $\iff$ "two-in, two-out"
- Ice-rule violation $(H_3O^+,\ OH^-) \iff$ **magnetic monopole**

Dense ice = classical **spin-1 spin ice**

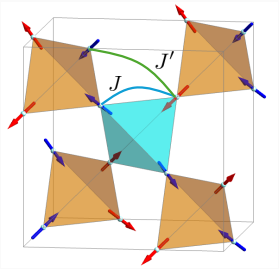
Harris *et al.*, PRL 79, 2554 (1997); Bramwell & Gingras, Science 294, 1495 (2001); Castelnovo, Moessner, Sondhi, Nature 451, 42 (2008); Pandey & Damle, arXiv:2512.11623

The effective model

$$H = J\sum_{\langle \ell\ell' \rangle} S^z_\ell S^z_{\ell'} + J'\sum_{\{\ell\ell'\}} S^z_\ell S^z_{\ell'} + \Delta \sum_\ell (S^z_\ell)^2$$

For $J' = 0$ and $\Delta = J + \mu$:

$$H = \frac{J}{2}\sum_{\boldsymbol{r}} \Big(\sum_{\ell \in \boldsymbol{r}} S^z_\ell \Big)^2 + \mu \sum_\ell (S^z_\ell)^2$$



- Δ: cost of off-center protons $\rightarrow$ drives ice X
- $J' > 0$: selects the polarized state $\rightarrow$ ice VIII
- monopole charge $Q_{\boldsymbol{r}} = \nabla \cdot \boldsymbol{S}_{\boldsymbol{r}} := \pm \sum_{\ell \in \boldsymbol{r}} S^z_\ell$ ($+$ on A, $-$ on B sites of the diamond lattice) costs $\frac{J}{2}Q^2$

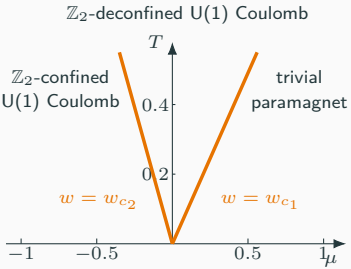
$w := e^{-\mu/T}$ (spin ± 1 vs. 0), $v := e^{-J/2T}$ (**monopoles**):

$$Z = \sum_{\{S^z_\ell\}} e^{-\beta H} = \sum_{\{Q_{\boldsymbol{r}}\}} \sum_{\{S^z_\ell\}} \prod_{\boldsymbol{r}} \delta_{\nabla \cdot \boldsymbol{S}_{\boldsymbol{r}}, Q_{\boldsymbol{r}}}\, v^{\sum_{\boldsymbol{r}} Q^2_{\boldsymbol{r}}}\, w^{\sum_\ell (S^z_\ell)^2}$$

Monopole-free limit $v = 0$: three phases, no symmetry breaking

$\boldsymbol{P} \coloneqq \frac{1}{L}\sum_{\ell=\langle \boldsymbol{r}_A \boldsymbol{r}_B\rangle}(\boldsymbol{r}_B - \boldsymbol{r}_A)\, S^z_\ell$ (net S^z flux through a plane; ice rule at $v = 0 \Rightarrow \boldsymbol{P} \in \mathbb{Z}^3$)

	$w < w_{c_1}$ trivial PM	$w_{c_1} < w < w_{c_2}$ $\mathbb{Z}_2$-deconfined	$w > w_{c_2}$ $\mathbb{Z}_2$-confined
ice	**ice X**	—	**ice VII**
emergent photon	gapped	gapless	gapless
flux sectors $\boldsymbol{P}$	$\mathbf{0}$	$\mathbb{Z}^3$	$(2\mathbb{Z})^3$
smallest free charge	—	$Q = 1$	$Q = 2$
transition	w_{c_1}: 3D XY		w_{c_2}: 3D Ising



$v = 0$ boundaries (Pandey & Damle):
$w = e^{-\mu/T} = w_c$, i.e. $\mu = -T \ln w_c$

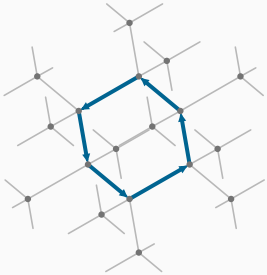
At $v = 0$, ice VII and ice X **are** distinct phases, separated by two topological transitions.

Duality I (small w): directed spin loops $\Rightarrow$ 3D XY model

$S^z = \pm 1$ links are unit currents in an $S^z = 0$ background. Gauss law at $v = 0 \Rightarrow$ **directed closed loops**. Write each constraint as $\delta_{\nabla\cdot \boldsymbol{S}_{\boldsymbol{r}},0} = \int_0^{2\pi} \frac{d\theta_{\boldsymbol{r}}}{2\pi}\, e^{-i\theta_{\boldsymbol{r}}\nabla\cdot \boldsymbol{S}_{\boldsymbol{r}}}$:

$$
\begin{aligned}
Z_{v=0} &= \textstyle\sum_{\{S^z_\ell\}} \prod_{\boldsymbol{r}} \delta_{\nabla\cdot \boldsymbol{S}_{\boldsymbol{r}},\,0}\; w^{\sum_\ell (S^z_\ell)^2} \\
&= \textstyle\int \mathcal{D}\theta \sum_{\{S^z_\ell\}} e^{-i\sum_{\boldsymbol{r}} \theta_{\boldsymbol{r}}\nabla\cdot \boldsymbol{S}_{\boldsymbol{r}}}\; w^{\sum_\ell (S^z_\ell)^2} \\
&= \textstyle\int \mathcal{D}\theta \prod_\ell \left[\sum_{S^z_\ell = 0,\pm 1} w^{(S^z_\ell)^2} e^{iS^z_\ell \nabla\theta_\ell}\right] \\
&= \textstyle\int \mathcal{D}\theta \prod_\ell \left[1 + 2w\cos\nabla\theta_\ell\right] \approx \int \mathcal{D}\theta\, e^{2w\sum_\ell \cos\nabla\theta_\ell},
\end{aligned}
$$

$\nabla\theta_\ell = \theta_{\boldsymbol{r}_B} - \theta_{\boldsymbol{r}_A}$ ($\sum_{\boldsymbol{r}} \theta_{\boldsymbol{r}}\nabla\cdot\boldsymbol{S}_{\boldsymbol{r}} = -\sum_\ell S^z_\ell \nabla\theta_\ell$): 3D XY, $K = 2w$.
T_c^{XY}(diamond) $= 1.30036(1) \Rightarrow w_{c_1} \simeq 0.385$ (MC: 0.3609)

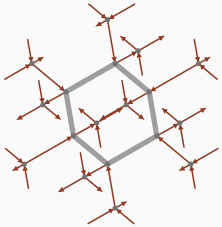


Loop proliferation at w_{c_1} = ordering of the dual XY model, $\langle e^{i\theta}\rangle \neq 0$

José, Kadanoff, Kirkpatrick, Nelson, PRB 16, 1217 (1977); Savit, Rev. Mod. Phys. 52, 453 (1980); S. Watanabe, Motome, H. Watanabe, J. Phys. Soc. Jpn. 95, 085002 (2026); arXiv:2603.03852

Duality II (large w): vacancy loops $\Rightarrow$ 3D Ising model

$S^z = 0$ links are unoriented vacancies in an $S^z = \pm 1$ ice background.
At $v = 0$ every site has an even number of them $\Rightarrow$ **undirected closed loops** C, each with $W(C)$ ice-rule backgrounds.



$$Z_{v=0} = \sum_{\{S^z_\ell\}} \prod_{\boldsymbol{r}} \delta_{\nabla\cdot \boldsymbol{S}_{\boldsymbol{r}},\,0}\; w^{\sum_\ell (S^z_\ell)^2} = \sum_C w^{2N-|C|}\, W(C)$$
$$= w^{2N} W(\emptyset) \sum_C \frac{W(C)}{W(\emptyset)}\, w^{-|C|} \overset{\text{Pauling}}{\approx} w^{2N} W(\emptyset) \sum_C \left(\tfrac{2}{3w}\right)^{|C|}$$

Pauling: $W(\emptyset) \approx 2^{2N}(3/8)^N$ and $W(C) \approx 2^{2N-|C|}(3/8)^{N-|C|}(1/2)^{|C|} \Rightarrow W(C)/W(\emptyset) \approx (2/3)^{|C|}$.
Compare with the **high-temperature expansion of the 3D Ising model** on the diamond lattice:

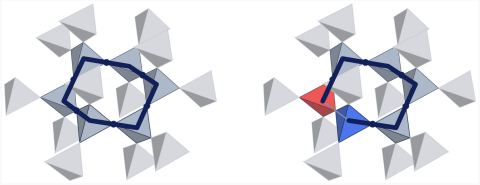
$$Z_{\text{Ising}} = \sum_{\{\sigma_{\boldsymbol{r}}\}} \prod_{\langle \boldsymbol{r}\boldsymbol{r}'\rangle} e^{K\sigma_{\boldsymbol{r}}\sigma_{\boldsymbol{r}'}} = \sum_{\{\sigma_{\boldsymbol{r}}\}} \prod_{\langle \boldsymbol{r}\boldsymbol{r}'\rangle} \left(\cosh K + \sigma_{\boldsymbol{r}}\sigma_{\boldsymbol{r}'} \sinh K\right) = (\cosh K)^{2N}\, 2^N \sum_C (\tanh K)^{|C|}$$

Ising critical coupling on the diamond lattice $K_c \simeq 0.3698 \Rightarrow w_{c_2} \simeq 1.884$ (MC: 1.8904)

Loop proliferation at w_{c_2} = ordering of the dual Ising model, $\langle \sigma_{\boldsymbol{r}} \rangle \neq 0$

Wegner (1971); Nahum *et al.*, PRL 107, 110601 (2011); Deng & Blöte, PRE 68, 036125 (2003); loop parities in the 3D toric-code limit of the Kitaev model: Nasu *et al.*, PRB 89, 115125 (2014)

Finite temperature: thermal monopoles open the loops



XY **picture** (small w)

$$Z \approx \int \mathcal{D}\theta \, e^{2w \sum_{\ell} \cos \nabla\theta_{\ell} \, + \, 2v \sum_{\boldsymbol{r}} \cos \theta_{\boldsymbol{r}}}$$

a uniform field $h = 2v$ on the dual XY spins

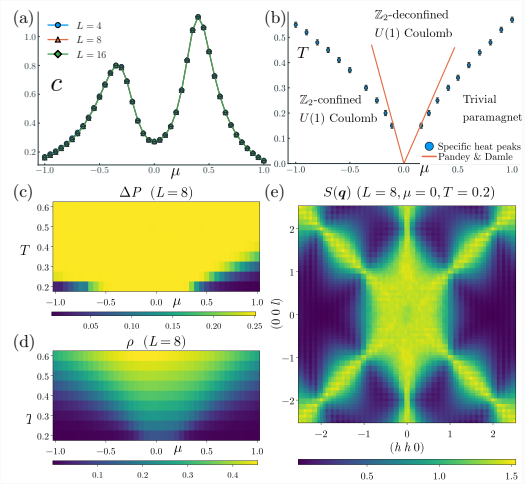
Ising picture (large w)

$$Z \approx \sum_{\{\sigma_{\boldsymbol{r}}\}} e^{K \sum_{\langle \boldsymbol{r}\boldsymbol{r}' \rangle} \sigma_{\boldsymbol{r}} \sigma_{\boldsymbol{r}'} + h \sum_{\boldsymbol{r}} \sigma_{\boldsymbol{r}}}$$

the Ising model in a field, $\tanh K = 2/(3w)$, $\tanh h = \sqrt{3}\, v$: the monopole fugacity is the field

Thermal monopoles = a **symmetry-breaking field** in the dual theory: every continuous transition becomes a **crossover** at any $T > 0$ (the field is relevant, $y_h > 0$).

Monte Carlo: rounded peaks, no quantized flux



(a) specific heat: $L = 4, 8, 16$ coincide, no divergence

(b) peak positions drift from the $v = 0$ boundaries: a **crossover diagram**

(c) global flux $\boldsymbol{P}$ leaks away from integers

(d) monopole density is finite

(e) pinch points persist at low T (long screening length)

At $T > 0$ the topological distinctions of the $v = 0$ model are gone.

Part II: summary

- Water and steam: one phase. Ice VII and ice X: one phase at $T > 0$.
- The $v = 0$ topological transitions are real, but **point-like monopoles open the loops** and act as a symmetry-breaking field in the dual theory $\Rightarrow$ crossover.
- Ice VIII is protected by broken symmetry; its boundary is sharp. For general S, only the first-order $S = \frac{3}{2}$ line can survive, ending at a critical end point.

For water, classification by spontaneous symmetry breaking is **complete**: wherever symmetry could not distinguish two states, a continuous path connected them.

To go beyond symmetry we need defects that **cannot** break. $\Rightarrow$ Part III

Part III: 3D toric code in fields — a thermal phase beyond symmetry

Open loops vs. closed loops: the principle

Spin ice / dense ice

the ice rule is **energetic**, cost $\frac{J}{2}Q^2$

thermal monopoles cut the loops

$\Rightarrow$ relevant dual ordering field

$\Rightarrow$ the transition becomes a **crossover**

3D toric code / $\mathbb{Z}_2$ gauge theory

Bianchi identity $\prod_{p\subset\partial c} B_p = \mathbf{1}$

flux loops are closed **at every** T

$\Rightarrow$ line tension vs. entropy

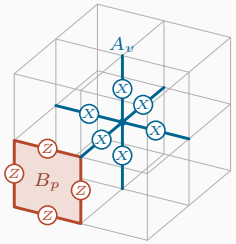
$\Rightarrow$ the thermal transition **survives**

Whether defect loops can break decides whether a thermal transition without symmetry breaking can exist.
In ice they break. In the 3D toric code they cannot — even when every microscopic symmetry is broken.

Castelnovo & Chamon, PRB 78, 155120 (2008); Nasu *et al.*, PRB 89, 115125 (2014); Fradkin & Shenker, PRD 19, 3682 (1979); S. Watanabe, Motome, H. Watanabe, arXiv:2603.03852

The model: 3D toric code with both fields

$$H = -J_e \sum_v A_v - J_m \sum_p B_p - h_x \sum_\ell X_\ell - h_z \sum_\ell Z_\ell, \qquad A_v = \prod_{\ell \ni v} X_\ell, \quad B_p = \prod_{\ell \in \partial p} Z_\ell$$



qubits on the links of a cubic lattice

$J_e = J_m = 1$; the two fields:

- h_z breaks the electric 1-form symmetry $U_e(\Sigma^*) = \prod_{\ell \perp \Sigma^*} X_\ell$
- h_x breaks the magnetic 2-form symmetry $U_m(C) = \prod_{\ell \in C} Z_\ell$
- both fields: **no microscopic higher-form symmetry remains**

Known limits:

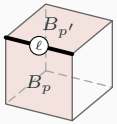
- $h = 0$: thermal transition $T_c^{(0)} \simeq 1.3133$ (3D Ising); TEE $= 2 \ln 2$ $(T = 0)$, $\ln 2$ $(0 < T < T_c^{(0)})$, 0 $(T > T_c^{(0)})$
- $T = 0$: first order at $h_x = 1$; continuous at $h_z \simeq 0.194$

Both microscopic symmetries explicitly broken. What, if anything, survives?

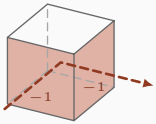
Wegner (1971); Castelnovo & Chamon, PRB 78, 155120 (2008); Reiss & Schmidt, SciPost Phys. 6, 078 (2019); Gaiotto, Kapustin, Seiberg, Willett, JHEP 02, 172 (2015); Bravyi, Hastings, Michalakis, J. Math. Phys. 51, 093512 (2010)

What survives: the closed-flux constraint

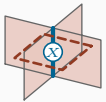
Bianchi identity on every cube c: $\prod_{p \subset \partial c} B_p = \mathbf{1}$



(a) each link ℓ of c lies in exactly two faces p, p', and $Z_\ell^2 = \mathbf{1}$



(b) so each cube has an *even* number of $B_p = -1$ faces (shaded): flux lines cannot end



(c) X_ℓ flips the four plaquettes around ℓ: the smallest flux loop

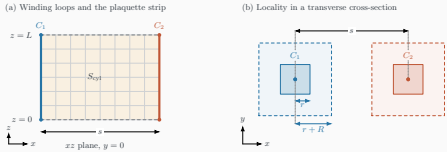
- plaquette defects ($B_p = -1$) have no endpoints: the dual edges piercing them form **closed flux loops**
- an **operator identity**: it holds at every T, h_x, h_z and for any perturbation — unlike the energetic ice rule, there is no endpoint mechanism
- low T: loops have tension; high T: loops proliferate

A constraint, not a symmetry, survives the explicit symmetry breaking — and it is exactly what a **decoder** needs.

The decoded Wilson-loop correlation f_W

$$f_W := \langle W_1 W_2 \rangle_{\mathcal{C}(\rho)} - \langle W_1 \rangle_{\mathcal{C}(\rho)} \langle W_2 \rangle_{\mathcal{C}(\rho)} = \langle \widetilde{W}_1 \widetilde{W}_2 \rangle_\rho - \langle \widetilde{W}_1 \rangle_\rho \langle \widetilde{W}_2 \rangle_\rho \qquad \widetilde{W}_i := \mathcal{C}^\dagger(W_i)$$

- two parallel winding loops C_1, C_2 at separation $s = \lfloor L/2 \rfloor$; $W_i = \prod_{\ell \in C_i} Z_\ell$, $W_1 W_2 = \prod_{p \in S_{\mathrm{cyl}}} B_p$
- a small flux loop near the strip flips the bare product $\Rightarrow$ first apply a **local recovery** $\mathcal{C}$: measure the syndrome η, flip the link set $\mathcal{R}(\eta)$ chosen by a fixed local decoder of depth $r = \lfloor \sqrt{L} \rfloor \ll s$



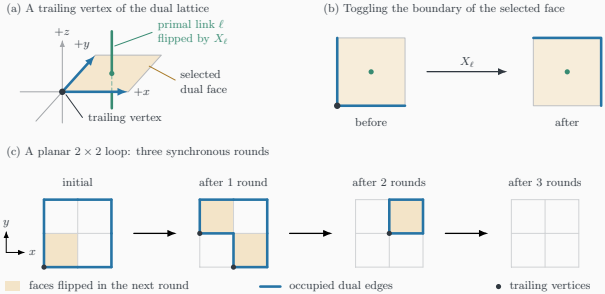
$$\mathcal{C}(\rho) := \sum_\eta X_{\mathcal{R}(\eta)} P_\eta \, \rho \, P_\eta X_{\mathcal{R}(\eta)}, \qquad P_\eta := \prod_p \frac{\mathbf{1} + \eta_p B_p}{2}, \qquad X_{\mathcal{R}(\eta)} := \prod_{\ell \in \mathcal{R}(\eta)} X_\ell$$

$$\langle W_i \rangle_{\mathcal{C}(\rho)} = \mathrm{Tr}\left[W_i \sum_\eta X_{\mathcal{R}(\eta)} P_\eta \, \rho \, P_\eta X_{\mathcal{R}(\eta)} \right] = \mathrm{Tr}\left[\widetilde{W}_i \, \rho \right] \quad \text{(cyclicity of the trace, } [P_\eta, W_i] = 0\text{)}$$

$$\widetilde{W}_i = \mathcal{C}^\dagger(W_i) = \sum_\eta P_\eta X_{\mathcal{R}(\eta)} W_i X_{\mathcal{R}(\eta)} P_\eta = W_i \sum_\eta (-1)^{|\mathcal{R}(\eta) \cap C_i|} P_\eta \quad (X_\ell W_i X_\ell = -W_i \text{ for } \ell \in C_i)$$

$\widetilde{W}_i$ is diagonal in the Z basis $\Rightarrow$ computable from QMC snapshots even at $h_x \neq 0$.

The fixed local decoder: Sweep



- a *trailing* dual vertex has occupied edges only in positive directions $\Rightarrow$ flip the link piercing the square they span
- synchronous rounds; r rounds erase any isolated component with $\leq 2(r+1)$ edges
- $w_i(z)$ depends only on spins within distance r of C_i: the two decoded observables live in **disjoint tubes**

Why the connected subtraction matters

$$D_{12} \coloneqq \langle \widetilde{W}_1 \widetilde{W}_2 \rangle_\rho, \quad M_i \coloneqq \langle \widetilde{W}_i \rangle_\rho, \quad f_W = D_{12} - M_1 M_2, \quad |f_W| \le \sqrt{(1-M_1^2)(1-M_2^2)} \le 1$$

shared random winding bit

$D_{12} = 1,\ M_i = 0 \Rightarrow f_W = 1$

e.g. the canonical $T = 0$ state $P_{0,L}/8$ (all 8 sectors)

pinned by polarization

$D_{12} = 1,\ M_1 M_2 = 1 \Rightarrow f_W = 0$

e.g. all $Z_\ell = +1$ (no syndrome, no correction)

f_W detects a **locally recoverable, unbiased, shared** bit — exactly what a product state cannot carry.

Theorem: nonzero f_W obstructs local preparation

Obstruction

Let $\sigma = \Phi(\pi)$ with π a product state and Φ a local circuit with product ancillas and causal radius R. If $2(R + r) < s$, then $f_W(\sigma) = 0$.

Proof. Dilate Φ to a unitary U on system + ancillas, $\Pi = \pi \otimes \pi_{\mathrm{anc}}$. The pulled-back observables $O_i = U^\dagger(\widetilde{W_i} \otimes \mathbf{1})U$ live in disjoint radius-$(R + r)$ tubes, so

$$D_{12}(\sigma) = \mathrm{Tr}(\Pi O_1 O_2) = \mathrm{Tr}(\Pi O_1)\,\mathrm{Tr}(\Pi O_2) = M_1(\sigma)M_2(\sigma). \qquad \square$$

- approximate preparation: $|f_W(\rho) - f_W(\sigma)| \le 3\|\rho - \sigma\|_1$
- Hamiltonian or Lindblad evolution for time $o(L)$: Lieb–Robinson tails $\Rightarrow f_W \to 0$

$\limsup_L |f_W| > 0 \Rightarrow$ **not** in the trivial phase of product states ($T = \infty$ and $h \to \infty$ included).

Lieb & Robinson, CMP 28, 251 (1972); Bravyi, Hastings, Verstraete, PRL 97, 050401 (2006); H. Watanabe, arXiv:2607.00134

An invariant of channel phases: $\mathcal{F}$

A fixed decoder can fail along a trivial path (uniform on-site rotation $U(t)$: $f_W(1/2) = 0$ although the state stays in the same phase). So **optimize** over binary observables near the loops and physical recoveries with total radius $\leq \epsilon L$:

$$F_{\epsilon,L}(\rho) \coloneqq \sup_{(W_1,W_2,\mathcal{C})} |f_W(\rho;\mathcal{C})|, \qquad \mathcal{F}(\boldsymbol{\rho}) \coloneqq \lim_{\epsilon\downarrow 0} \limsup_{L\to\infty} F_{\epsilon,L}(\rho_L)$$

Phase invariant

$\mathcal{F}$ is nonincreasing under local channels with product ancillas and radius $o(L)$. Hence $\boldsymbol{\rho} \sim \boldsymbol{\sigma} \Rightarrow \mathcal{F}(\boldsymbol{\rho}) = \mathcal{F}(\boldsymbol{\sigma})$.

- $\mathcal{F} = 1$ for the canonical $T = 0$ field-free state; $\mathcal{F} = 0$ for every product state
- lower bound from the fixed decoder: $\mathcal{F} \geq \limsup_L |f_W|$, so $f_W \to 1$ implies $\mathcal{F} = 1$

$\mathcal{F}$ is constant on every two-way local-channel phase, and the measured $f_W \to 1$ forces $\mathcal{F} = 1$.

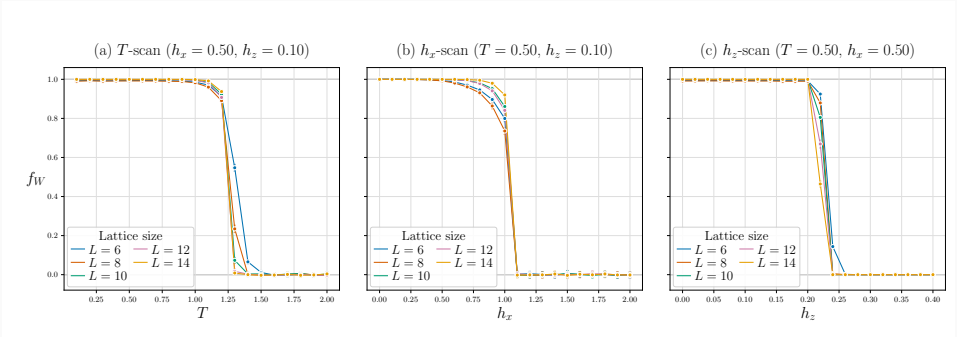
Why $\mathcal{F}$ is a phase invariant: proof

Let Φ_L be a local circuit with product ancillas and causal radius $R_L = o(L)$, and let $e_L := \|\sigma_L - \Phi_L(\rho_L)\|_1 \to 0$.

1. **Pull the candidate back.** For any admissible $(W_1, W_2, \mathcal{C})$ for σ_L: $f_W\big(\Phi_L(\rho_L); \mathcal{C}\big) = f_W\big(\rho_L; \mathcal{C} \circ \Phi_L\big)$, and $\mathcal{C} \circ \Phi_L$ is again a physical circuit with product ancillas whose radius grows by at most R_L.
2. **Conversion error.** $|f_W(\sigma_L; \mathcal{C}) - f_W(\Phi_L(\rho_L); \mathcal{C})| \le 3e_L$: covariances are trace-norm stable and $\mathcal{C}$ is contractive.
3. **Radius budget.** Fix a small $\epsilon > 0$; for large L, $R_L/L < \epsilon$, so
$$F_{\epsilon, L}(\sigma_L) \le F_{\epsilon + R_L/L,\, L}(\rho_L) + 3e_L \le F_{2\epsilon, L}(\rho_L) + 3e_L.$$
4. **Limits.** $\limsup_{L\to\infty}$ first (absorbs any sublinear enlargement), then $\epsilon \downarrow 0$: $\mathcal{F}(\boldsymbol{\sigma}) \le \mathcal{F}(\boldsymbol{\rho})$. A reverse conversion gives $\ge$. □

One-way local conversion cannot increase $\mathcal{F}$; two-way conversion fixes it. Product inputs give $\mathcal{F} = 0$ (disjoint backward cones), so $\mathcal{F} > 0$ excludes the trivial phase.

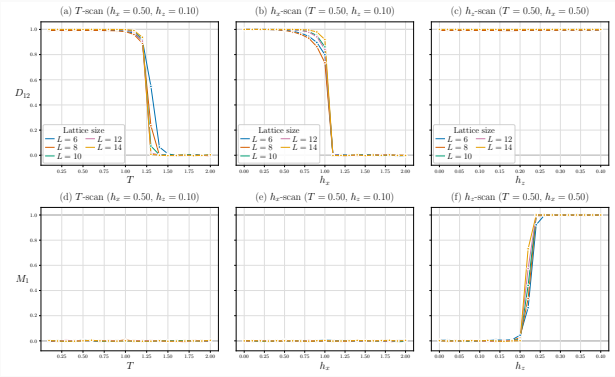
Quantum Monte Carlo: f_W along three cuts



- worldline QMC, equal-time Z snapshots, $L = 6, 8, 10, 12, 14$, fixed Sweep depth $r = \lfloor \sqrt{L} \rfloor$
- (a) T at $(h_x, h_z) = (0.5, 0.1)$; (b) h_x at $(T, h_z) = (0.5, 0.1)$; (c) h_z at $(T, h_x) = (0.5, 0.5)$

$f_W \simeq 1$ at low T and weak fields; $f_W \to 0$ beyond, sharpening with L.

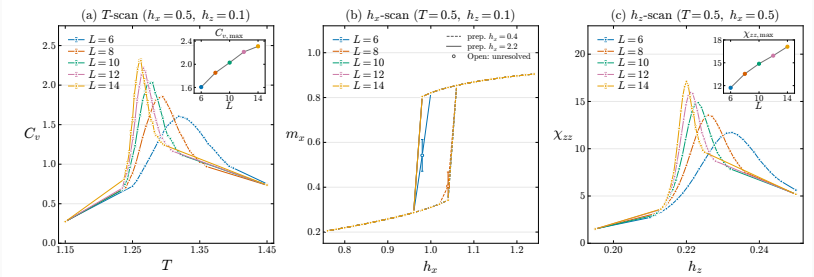
Two different ways to lose the bit



T and h_x scans: D_{12} drops while $M_i \simeq 0$
$\Rightarrow$ the two loop values stop agreeing

h_z scan: $D_{12} \simeq 1$ but $M_i \to 1 \Rightarrow$ polarization;
the subtraction is essential

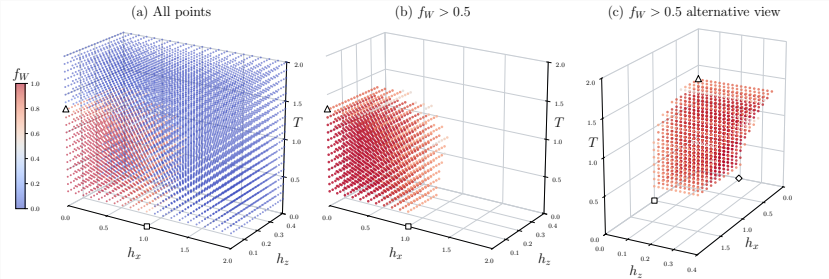
Thermodynamic signatures on the same cuts



- (a) the specific-heat peak and (c) the longitudinal-susceptibility peak grow with L where f_W drops
- (b) transverse polarization: preparation-dependent branches near the drop of f_W

The loss of f_W is accompanied by thermodynamic singularities.

The region with $\mathcal{F} = 1$



- $L = 8$ survey over (h_x, h_z, T): 8379 points; (b,c) the points with $f_W > 0.5$
- markers: known transitions at $(h_x, h_z, T) = (1, 0, 0),\ (0, 0.194, 0),\ (0, 0, 1.3133)$

An extended region at $T > 0,\ h_x, h_z > 0$ with $\mathcal{F} = 1$: a Gibbs-state phase **beyond microscopic symmetry breaking**.

Part III: summary, what $\mathcal{F}$ is not, and what is still missing

- $\mathcal{F} = 1$ and $\mathcal{F} = 0$ are **different phases** under two-way local channels; numerics give $f_W \to 1$, hence $\mathcal{F} = 1$, in a finite region of (T, h_x, h_z) with no microscopic higher-form symmetry.
- $\mathcal{F}$ is not quantized in general ($\mathcal{F} = 4p(1-p)$ for a mixture of two ferromagnets), $\mathcal{F} = 0$ does not prove triviality, and the bit is classical (three logical Z parities): no quantum memory is claimed.

Left open here	$\Rightarrow$	**Proven for general $\mathbb{Z}_2$ cell models**
two uniform fields h_x, h_z of one model; evidence by QMC	$\Rightarrow$	**arbitrary** weak local perturbations, any (d, m); theorems
thermodynamic connectivity: is f regular inside the $\mathcal{F} = 1$ region?	$\Rightarrow$	joint analyticity of f at low (and high) T
channel connectivity: only the one-way obstruction and $\mathcal{F} \geq \limsup \lvert f_W \rvert$; no two-way channel to a reference state	$\Rightarrow$	$\boldsymbol{\rho}_{\beta,g} \sim_{\mathrm{LC}} \boldsymbol{\rho}_{\mathrm{flat}}$ **both ways**: the whole domain is one phase, $\mathcal{F}_m = 1$

Closed flux loops + local decoding $\Rightarrow$ a thermal phase invariant beyond symmetry. The $\mathbb{Z}_2$ cell models of Sengoku, Cao, HW and Xun, Cao, HW turn it into theorems.

H. Watanabe, arXiv:2607.00134; cell models: Sengoku, Cao, H. Watanabe, arXiv:2610.27938; Xun, Cao, H. Watanabe, arXiv:2610.27939; cf. Hastings, PRL 107, 210501 (2011); Castelnovo & Chamon (2008); Stahl, Placke, Khemani, Li, PRE 114, 034126 (2026)

Conclusion

Summary

- **Question I**: time crystals (long-range order in time) do not exist in equilibrium with short-range interactions
- **Question II**: yes — at $T > 0$, whether the defect loops can break decides whether a distinction survives
- **Not special to the quantum toric code**: for classical $\mathbb{Z}_2$ cell models, stability of the $\mathcal{F} = 1$ phase under arbitrary weak local perturbations is a **theorem** (Sengoku, Cao, HW; Xun, Cao, HW)

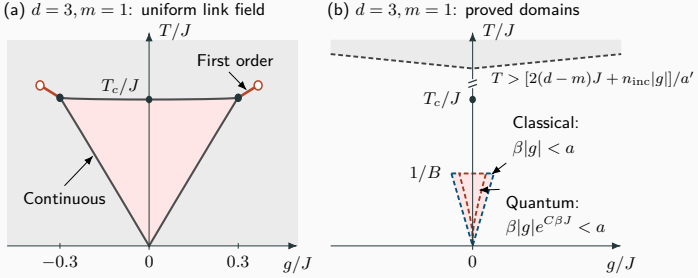
	spin ice / dense ice	3D toric code in fields
defect constraint	energetic ice rule	Bianchi identity
can loops break?	**yes** (monopoles)	**no**
sector label	global flux (lost at $T > 0$)	winding Wilson loops
fate at $T > 0$	crossover	$\mathcal{F} = 1$ region (QMC)
classification	symmetry suffices	beyond symmetry

Beyond symmetry breaking at $T > 0$: a **nonlocal sector label** that bounded regions cannot read, carried by **closed extended defects** that local operations can clean up.

S. Sengoku, W. Cao, H. Watanabe, arXiv:2610.27938; G. Xun, W. Cao, H. Watanabe, arXiv:2610.27939; H. Watanabe, arXiv:2607.00134

Beyond the toric code: classical $\mathbb{Z}_2$ cell models, the $(d, m) = (3, 1)$ case

The $(d, m) = (3, 1)$ cell model is the classical 3D $\mathbb{Z}_2$ gauge theory (link spins, plaquette interactions), the classical core of the 3D toric code. (a) Schematic phase diagram in a uniform link field g: the deconfined region survives $g \neq 0$ with no symmetry pinning its boundaries. (b) Where the theorems apply.



Classical wedge, narrower quantum window, high-T domain reaching $T = \infty$.
Dashed lines are bounds, not phase boundaries; the white region is open.

(a): Somoza, Serna, Nahum, PRX 11, 041008 (2021); Bonati, Pelissetto, Vicari, PRB 105, 165138 (2022); Fradkin & Shenker (1979). (b): S. Sengoku, W. Cao, H. Watanabe, arXiv:2610.27938 (Fig. 2); channels: G. Xun, W. Cao, H. Watanabe, arXiv:2610.27939

Outlook

- **Thermal phase separation**: prove that a thermodynamic singularity separates the two analytic domains
- **Sharp boundaries**: where does $\mathcal{F}$ jump?
  relation between channel phases and free-energy analyticity
- **Experiment**: realizing the 3D toric code and its thermal phase
- **Beyond**: searching for other finite-temperature topological phases

Thank you!

I. H. Watanabe, M. Oshikawa, PRL 114, 251603 (2015); H. Watanabe, M. Oshikawa, T. Koma, J. Stat. Phys. 178, 926 (2020)

II. S. Watanabe, Y. Motome, H. Watanabe, PRB 114, 094108 (2026); arXiv:2603.03852; PRX 16, 041016 (2026); JPSJ 95, 085002 (2026)

III. H. Watanabe, arXiv:2607.00134; S. Sengoku, W. Cao, H. Watanabe, arXiv:2610.27938; G. Xun, W. Cao, H. Watanabe, arXiv:2610.27939